

## Team – Circuit Breakers

| Sr. No. | Team Member      | Department |
|---------|------------------|------------|
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### I. Problem Statement & Observation:

*Patients, especially elderly individuals and those managing multiple medications, require a reliable, affordable, and automated medicine dispensing system because manually remembering medication schedules often leads to missed or incorrect doses. Existing solutions are expensive, bulky, or lack personalized scheduling, making them impractical for everyday home use.*

Major Observations -

- Patients frequently forgot to take medicines at the prescribed time.
- Caregivers had to manually remind patients multiple times a day.
- Missing or incorrect medication doses affected treatment adherence.
- Managing multiple medicines with different schedules caused confusion.
- Existing automated medicine dispensers were expensive and inaccessible for many users.
- Conventional pill organizers provided storage but did not offer reminders or automatic dispensing.
- Patients often forgot whether they had already taken a scheduled dose.
- Family members could not easily monitor whether medicines were taken on time.
- Elderly users preferred a simple system with audio and visual reminders instead of mobile app notifications.
- A low-cost, automated medicine dispenser with scheduled dispensing and confirmation of medicine retrieval would significantly improve medication adherence and patient independence.

### II. Solution & Design:

The proposed **Smart Medicine Dispenser** is a compact, low-cost automated medication management system designed to assist patients in taking their medicines on time. The system stores up to three different medicines, dispenses them according to a predefined schedule, and provides both audio and visual reminders to ensure timely medication intake. Unlike many commercial automated dispensers, the proposed solution focuses on affordability, ease of use, and reliability, making it suitable for home use, especially for elderly patients and individuals managing multiple medications.

The operation follows these steps:

1. A medicine schedule is registered through the ESP8266 and transmitted to the Arduino Mega via Serial3.
2. The Arduino Mega stores the medicine name, dosage schedule, dispensing times, and treatment duration.

3. The DS3231 Real-Time Clock (RTC) continuously tracks the current date and time.
4. At the scheduled time, the Arduino Mega compares the current time with the stored medication schedule.
5. The corresponding stepper motor rotates to dispense the prescribed medicine from the appropriate compartment.
6. The DFPlayer Mini plays an audio reminder while the LCD displays the medicine information and dispensing status.
7. The IR sensor detects whether the medicine has been collected by the patient.
8. Once the medicine is collected, the system records the dose as completed and waits for the next scheduled dispensing time.
9. The process repeats automatically until the prescribed treatment duration is completed, after which the medicine schedule is removed from the system.

The project began by studying the challenges faced by patients and caregivers in maintaining medication schedules through observations and discussions. The team identified the need for an affordable, automated medicine dispenser capable of handling multiple medicines with minimal user intervention. Suitable hardware components were selected, including an Arduino Mega, ESP32, RTC module, stepper motors, IR sensor, DFPlayer Mini, and LCD display. The hardware and software were then integrated, and the complete prototype was developed and tested to validate accurate scheduling, reliable dispensing, and user-friendly operation.

### III. Components & Technical Planning:

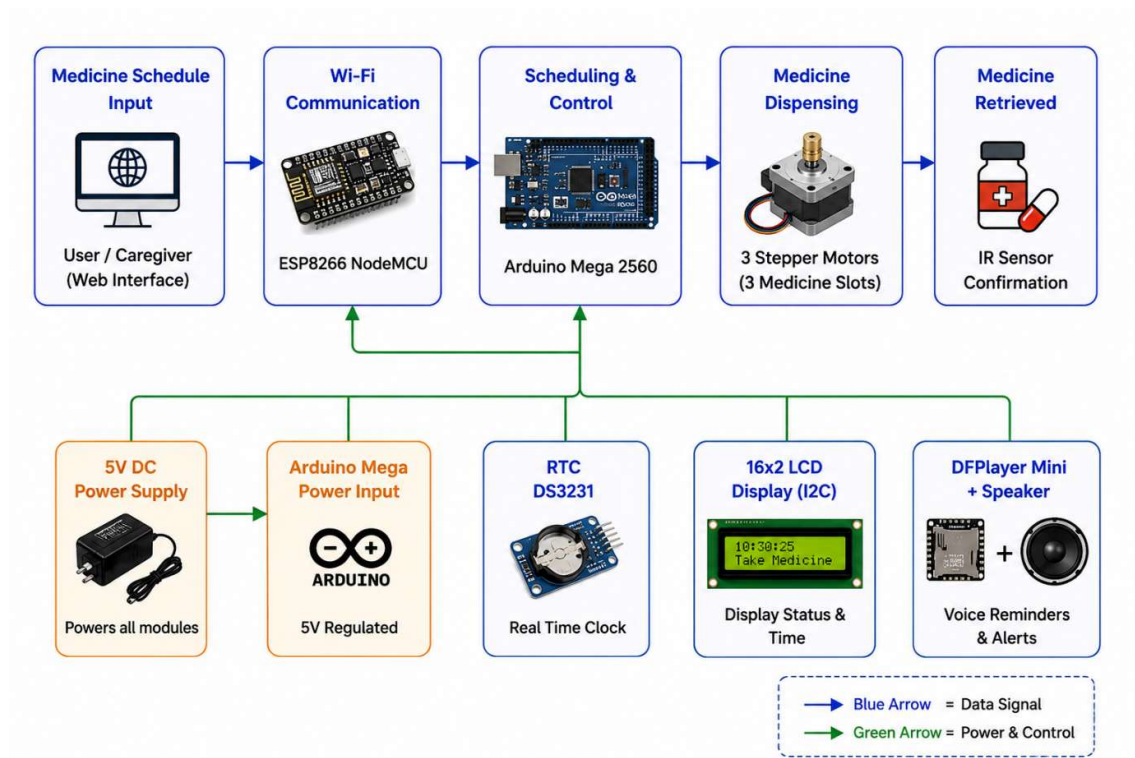
| Sr. No. | Component                                  | Purpose                                                                                                                                                                      |
|---------|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1       | <b>Arduino Mega 2560</b>                   | Acts as the main controller, stores medicine schedules, processes dispensing logic, controls the stepper motors, LCD, RTC, DFPlayer Mini, and communicates with the ESP8266. |
| 2       | <b>ESP8266 NodeMCU</b>                     | Provides Wi-Fi connectivity, hosts the web interface for medicine registration, and transmits medicine schedules to the Arduino Mega through Serial3 communication.          |
| 3       | <b>DS3231 RTC Module</b>                   | Maintains accurate date and time, allowing medicines to be dispensed precisely according to the prescribed schedule, even after power interruptions.                         |
| 4       | <b>28BYJ-48 Stepper Motors (×3)</b>        | Rotate the medicine compartments to dispense the correct medicine from each of the three slots at the scheduled time.                                                        |
| 5       | <b>ULN2003 Stepper Driver Modules (×3)</b> | Drive the 28BYJ-48 stepper motors by providing the required current and control interface from the Arduino Mega.                                                             |
| 6       | <b>16×2 I2C LCD Display</b>                | Displays the current time, medicine name, dispensing status, and system notifications for the user.                                                                          |

|    |                                     |                                                                                                                                         |
|----|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| 7  | <b>DFPlayer Mini</b>                | Plays pre-recorded voice reminders and audio notifications to alert the patient when it is time to take the prescribed medicine.        |
| 8  | <b>Speaker</b>                      | Outputs the voice reminders and alert tones from the DFPlayer Mini.                                                                     |
| 9  | <b>IR Sensor Module</b>             | Detects whether the dispensed medicine has been collected by the patient, allowing the system to confirm successful medicine retrieval. |
| 10 | <b>5V DC Power Supply / Adapter</b> | Provides a stable power supply to the Arduino Mega, ESP8266, and all connected modules.                                                 |

#### IV. Challenges & Solutions:

The project faced several challenges during both the hardware and software development phases. Mechanically, designing a reliable 3D-printed medicine dispensing mechanism required multiple iterations to achieve smooth rotation, accurate alignment of the medicine compartments, and consistent dispensing without pills getting stuck or jammed. Adjustments to the compartment dimensions, clearances, and stepper motor coupling were necessary to improve reliability and ensure proper operation.

On the electronics and software side, challenges included maintaining accurate medicine dispensing schedules, ensuring reliable communication between the ESP8266 and Arduino Mega, synchronizing the RTC with the dispensing logic, detecting successful medicine retrieval using the IR sensor, and integrating multiple hardware modules into a single system. These challenges were addressed by using the DS3231 RTC for precise timekeeping, implementing robust serial communication between the ESP8266 and Arduino Mega, refining the 3D-printed design through repeated prototyping, and adopting a modular software architecture for easier integration and maintenance. Careful component selection, extensive testing, and effective teamwork helped overcome budget, mechanical, and development time constraints, resulting in a reliable, affordable, and user-friendly automated medic



ne dispenser.